

Team Spending Versus Performance in Professional Sports

Is Increased Payroll and Transfer Spending Associated with Greater Success?

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Introduction

Professional sports are a billion-dollar industry. Each season, athletes from many leagues all around the world earn millions of dollars for their performance, leading one to think that compensation is based on athletic prowess, and therefore higher-spending teams are more successful. This relationship, however, is not always clear. This project seeks to quantitatively answer the question: do teams that have higher payroll and transfer spending achieve greater success, as measured by their regular season and postseason performance? Multiple data visualizations highlight unique relationships across the four dimensions in numerous sports.

Methodology

The backbone of the methodology of the project is Pearson's correlation to inspect how closely the features follow each other, as well as empirical examination of scatter plots of each of the types of spending and success metrics. Throughout the entire project, I have used Python. Specifically, once I had found a trustworthy website with the information that I need (see the list of sources in references), I have used the *requests* and *BeautifulSoup* libraries to scrape information from it and put it into a suitable format. This process was used for all leagues and all data except for postseason performance, which, due to a lack of available easily scrapable resources, was researched using a variety of sources manually. With the help of such libraries as *pandas* and *numpy*, I have then combined all the data into a single table and visualized the findings using *matplotlib*.

In order to calculate spending, two metrics were used:

- Payroll spending denotes how much a team has spent either on average annually or in total over the past 5 years on the salaries of players, managers, and other staff.
- Transfer spending denotes how much a team has spent either on average annually or in total over the past 5 years on purchasing players from other teams. This metric was only used for soccer, since in North American sports leagues players leave their teams much more rarely, and when they do, they usually do it as free agents or are traded for draft picks and other players. Unlike payroll spending, transfer spending can be negative, indicating that a team has made net profit by selling more than they bought.

Note that for both of these metrics, inflation was not taken into account. Additionally, the unit used for both of these metrics in all tables and graphs is millions USD. These numbers are accurate as of 7/7/2021.

Team success was also measured using multiple metrics:

- Season points, used in soccer and hockey, is the primary measure of regular season performance for these two sports. In soccer, a team gets no points for losing, 1 point for a draw, and 3 points for a win. In hockey, a team gets no points for losing in regular time, 1 point for losing in overtime, and 2 points for a win.
- The winning percentage (referred to as WPCT throughout the project) is used to determine regular season success in football, basketball, and baseball. It is simply the percentage of regular season games that a team has won.
- Playoff points calculate postseason performance for all sports in knockout competitions, including playoffs, FA Cup, UEFA Champions League, and similar competitions. Playoff points are based on how far teams have gone in these tournaments, but the exact method of calculating this metric was slightly different from league to league:
 - EPL:
 - UEFA Champions League: 1 point for reaching round of 16, 2 for reaching quarterfinals, 3 for reaching semifinals, 4 for reaching finals, 5 for winning.
 - FA Cup: 1 point for reaching 5th round proper, 2 for reaching quarterfinals, 3 for reaching semifinals, 4 for reaching finals, 5 for winning.
 - UEFA Europa League: 1 point for reaching quarterfinals, 2 for reaching semifinals, 3 for reaching finals, 4 for winning.
 - EFL Cup: 1 point for reaching semifinals, 2 for reaching finals, 3 for winning.
 - Serie A:
 - UEFA Champions League: 1 point for reaching round of 16, 2 for reaching quarterfinals, 3 for reaching semifinals, 4 for reaching finals, 5 for winning.
 - Coppa Italia: 1 point for reaching round of 16, 2 for reaching quarterfinals, 3 for reaching semifinals, 4 for reaching finals, 5 for winning.
 - UEFA Europa League: 1 point for reaching quarterfinals, 2 for reaching semifinals, 3 for reaching finals, 4 for winning.
 - MLS:
 - CONCACAF Champions League: 1 point for reaching round of 16 (if available that year), 2 for reaching quarterfinals, 3 for reaching semifinals, 4 for reaching the final, 5 for winning.
 - US Open Cup: 1 point for reaching round of 16, 2 for reaching quarterfinals, 3 for reaching semifinals, 4 for reaching the final, 5 for winning.
 - MLS Cup: 1 point for reaching the First Round, 2 for reaching Conference Semifinals, 3 for reaching Conference Finals, 4 for reaching the final, 5 for winning.
 - NFL and NBA:

- 1 point for reaching the Wildcard Round, 2 for reaching Divisional Playoffs, 3 for reaching Conference Championship, 4 for reaching the final, 5 for winning.
- NHL:
 - 1 point for reaching First Round, 2 for reaching Second Round, 3 for reaching Stanley Cup semifinals, 4 for reaching Stanley Cup finals, 5 for winning.
- MLB:
 - 1 point for reaching Wildcard Series, 2 for reaching Division Series, 3 for reaching League Championship Series, 4 for reaching World Series, 5 for winning.

A couple things are worth noting about this system. First, these points add up through rounds; for example, a Superbowl winner will get 15 points (1+2+3+4+5 after winning all of the previous games), not just 5. Second of all, teams that skip a stage (e.g. NFL Divisional Winners that got to the divisional playoffs without playing in the wildcard round) get treated as if they won the stage they skipped and get +3 points rather than +2.

Data

Soccer

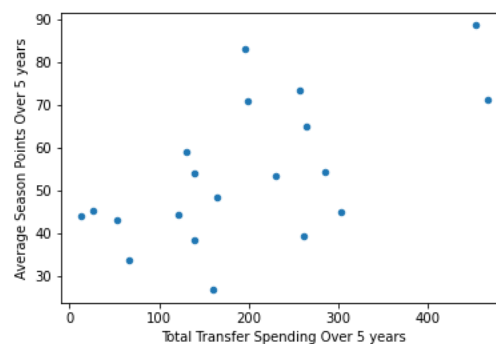
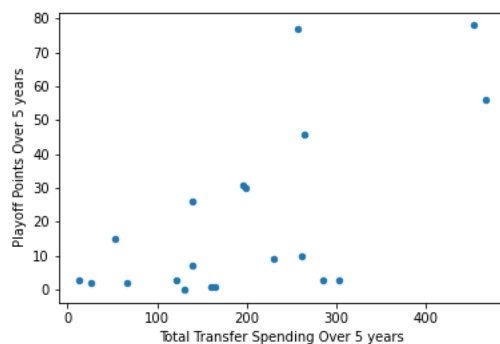
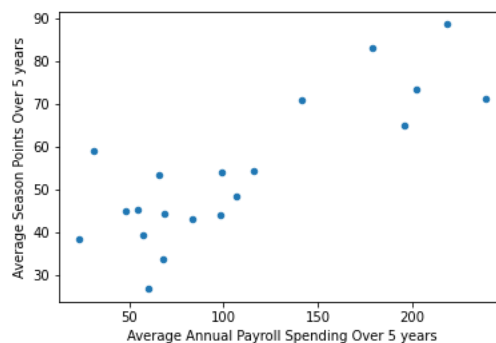
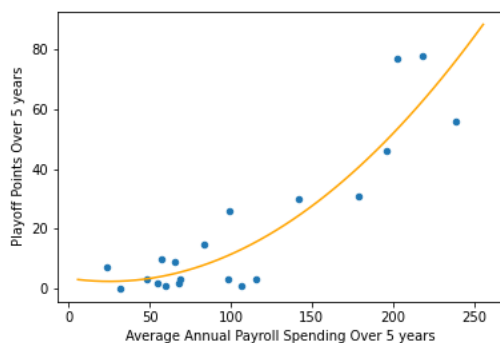
English Premier League

Data Table

	Total Transfer Spending Over 5 years	Average Annual Payroll Spending Over 5 years	Average Season Points Over 5 years	Playoff Points Over 5 years
Arsenal FC	263.67	195.71	65	46
Aston Villa	302.94	48.09	45	3
Brighton & Hove Albion	260.75	57.49	39.5	10
Burnley FC	26.99	54.74	45.4	2
Chelsea FC	256.72	202.37	73.6	77
Crystal Palace	13.28	98.27	44.2	3
Everton FC	284.81	115.82	54.4	3
Fulham FC	160.8	59.96	27	1
Leeds United	130.03	31.45	59	0
Leicester City	139.78	99.2	54.2	26
Liverpool FC	196.36	178.85	83.2	31
Manchester City	453.76	218.33	88.6	78
Manchester United	465.79	238.64	71.2	56

Newcastle United	120.96	68.44	44.5	3
Sheffield United FC	140	23.61	38.5	7
Southampton FC	53.35	83.8	43.2	15
Tottenham Hotspur	198.61	141.64	71	30
West Bromwich Albion FC	66.9	67.81	34	2
West Ham United	165.04	106.92	48.6	1
Wolverhampton Wanderers	230.05	65.63	53.67	9

Graphs



Pearson's correlation

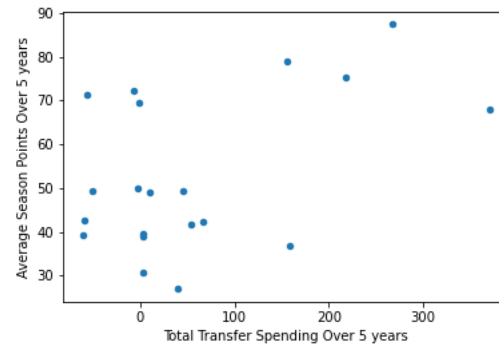
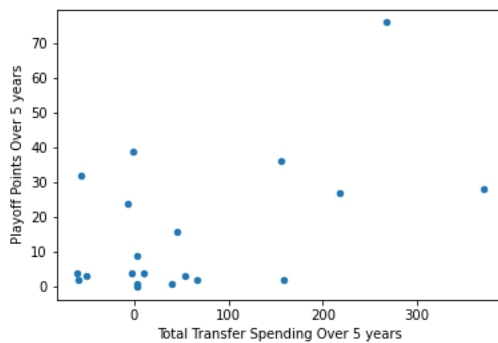
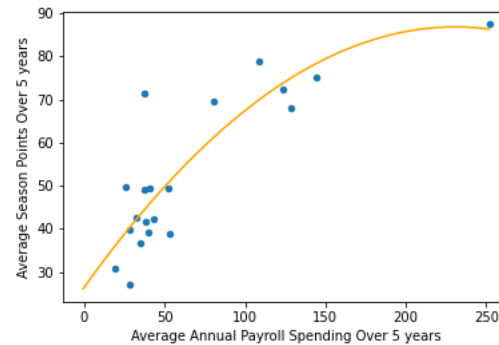
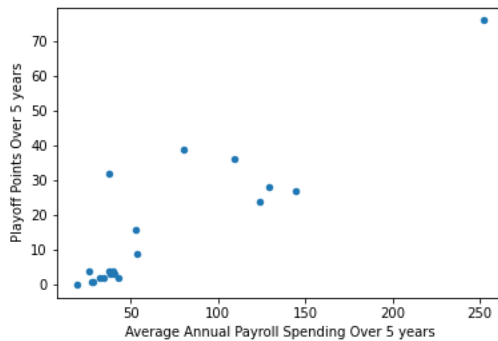
	Playoff Points	Season Points
Payroll Spending	0.882	0.822
Transfer Spending	0.706	0.601

Serie A

Data Table

	Total Transfer Spending Over 5 years	Average Annual Payroll Spending Over 5 years	Average Season Points Over 5 years	Playoff Points Over 5 years
AC Milan	370.04	129	68	28
ACF Fiorentina	45.95	52.6	49.4	16
AS Roma	-7.43	123.76	72.4	24
Atalanta BC	-56.24	37.27	71.4	32
Benevento	40.25	27.96	27	1
Bologna FC 1909	66.42	43.08	42.4	2
Cagliari Calcio	54.34	38.32	41.8	3
Crotone	3	19.44	30.67	0
Genoa CFC	-60.41	39.51	39.2	4
Hellas Verona	2.38	28.16	39.67	1
Inter Milan	218.22	144.23	75.2	27
Juventus FC	267.72	251.8	87.4	76
Parma	158	34.91	36.67	2
SS Lazio	-1.39	80.44	69.4	39
SSC Napoli	155.05	109	79	36
Spezia Calcio	-2.26	26.18	49.8	4
Torino FC	2.53	53.55	39	9
UC Sampdoria	-50.35	40.46	49.4	3
US Sassuolo	9.68	37.37	49	4
Udinese Calcio	-59.04	32.37	42.6	2

Graphs



Pearson's Correlation

	Playoff Points	Season Points
Payroll Spending	0.888	0.817
Transfer Spending	0.528	0.476

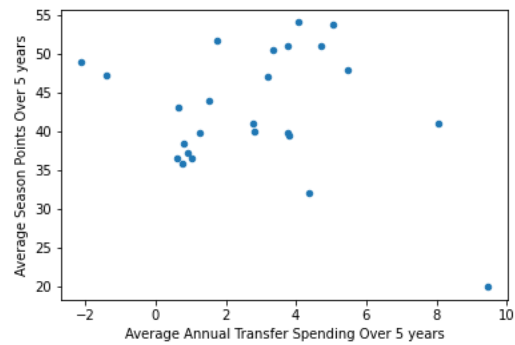
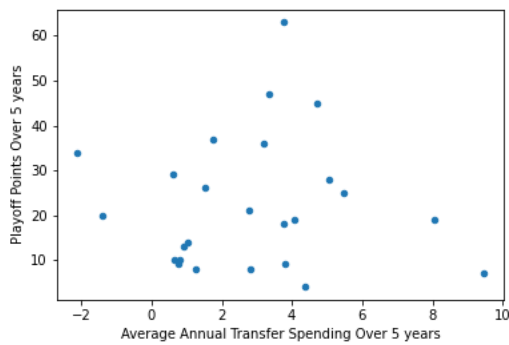
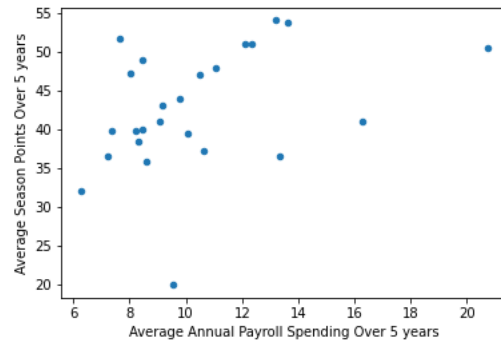
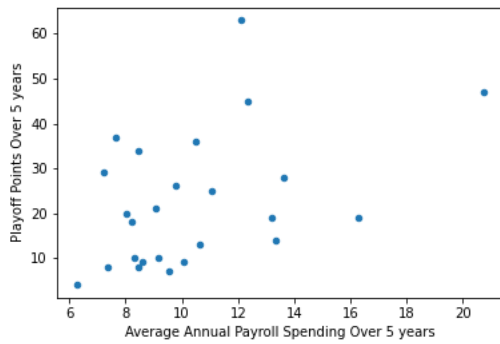
Major League Soccer

Note: Inter Miami was dropped since it is an outlier that creates an illusion of correlation when there's none. It is a relatively new club, which has spent 17 million on transfers and 14 million on payroll on average (much more than other teams), while at the same time having no playoff points and really low season points (26). As such, it created a moderate negative correlation between spending and success, when in reality it is either moderately positive (in the case of payroll) or nonexistent (in the case of transfer spending).

Data Table

	Average Annual Transfer Spending Over 5 years	Average Annual Payroll Spending Over 5 years	Average Season Points Over 5 years	Playoff Points Over 5 years
Atlanta United FC	4.73	12.33	51	45
CF Montreal	3.8	10.08	39.4	9
Chicago Fire	1.05	13.32	36.6	14
Colorado Rapids	0.82	8.29	38.4	10
Columbus Crew	1.51	9.77	44	26
D.C. United	2.8	8.45	40	8
FC Cincinnati	9.44	9.52	20	7
FC Dallas	-2.11	8.46	49	34
Houston Dynamo	0.63	7.22	36.6	29
LA Galaxy	8.07	16.3	41	19
Los Angeles FC	5.07	13.62	53.67	28
Minnesota United FC	3.78	8.22	39.75	18
Nashville SC	4.37	6.28	32	4
New England Revolution	2.78	9.09	41	21
New York City FC	4.05	13.2	54	19
New York Red Bulls	1.76	7.64	51.6	37
Orlando City	0.91	10.64	37.2	13
Philadelphia Union	-1.4	8.02	47.2	20
Portland Timbers	5.49	11.04	47.8	25
Real Salt Lake	0.67	9.17	43	10
San Jose Earthquakes	0.78	8.58	35.8	9
Seattle Sounders FC	3.76	12.11	51	63
Sporting Kansas City	3.2	10.49	47	36
Toronto FC	3.36	20.72	50.4	47
Vancouver Whitecaps FC	1.25	7.38	39.8	8

Graphs



Pearson's Correlation

	Playoff Points	Season Points
Payroll Spending	0.431	0.382
Transfer Spending	0.002	-0.202

Other sports

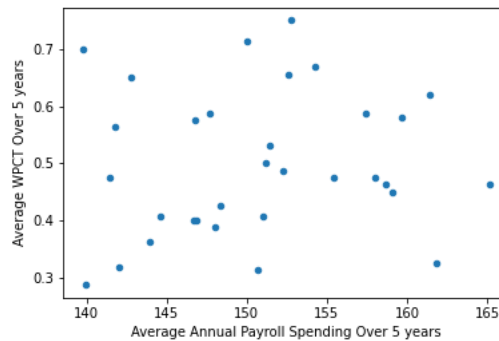
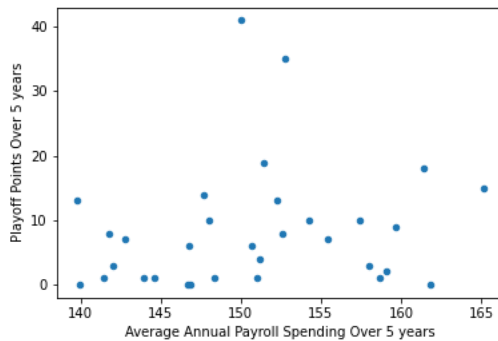
National Football League

Data Table

	Average Annual Payroll Spending Over 5 years	Average WPCT Over 5 years	Playoff Points Over 5 years
Arizona Cardinals	146.9	0.4	0
Atlanta Falcons	152.28	0.488	13
Baltimore Ravens	142.77	0.65	7

Buffalo Bills	141.76	0.563	8
Carolina Panthers	148.37	0.425	1
Chicago Bears	159.12	0.45	2
Cincinnati Bengals	161.81	0.325	0
Cleveland Browns	142.07	0.319	3
Dallas Cowboys	146.77	0.575	6
Denver Broncos	146.68	0.4	0
Detroit Lions	144.62	0.407	1
Green Bay Packers	161.41	0.619	18
Houston Texans	155.44	0.475	7
Indianapolis Colts	151.15	0.5	4
Jacksonville Jaguars	150.72	0.313	6
Kansas City Chiefs	152.76	0.75	35
Las Vegas Raiders	158.67	0.463	1
Los Angeles Chargers	157.97	0.475	3
Los Angeles Rams	147.69	0.588	14
Miami Dolphins	141.49	0.475	1
Minnesota Vikings	159.68	0.581	9
New England Patriots	149.99	0.713	41
New Orleans Saints	139.81	0.7	13
New York Giants	143.93	0.363	1
New York Jets	139.98	0.288	0
Philadelphia Eagles	151.45	0.532	19
Pittsburgh Steelers	154.25	0.669	10
San Francisco 49ers	148.05	0.388	10
Seattle Seahawks	152.58	0.656	8
Tampa Bay Buccaneers	165.12	0.463	15
Tennessee Titans	157.44	0.588	10
Washington Football Team	151	0.407	1

Graphs



Pearson's Correlation

	Playoff Points	WPCT
Payroll Spending	0.168	0.098

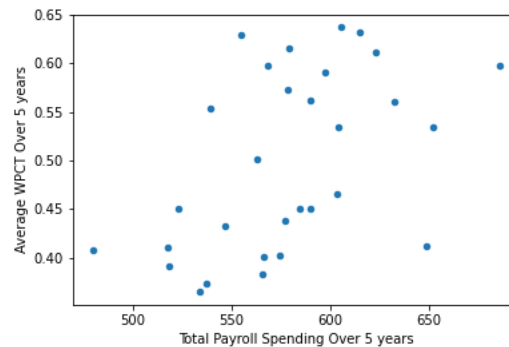
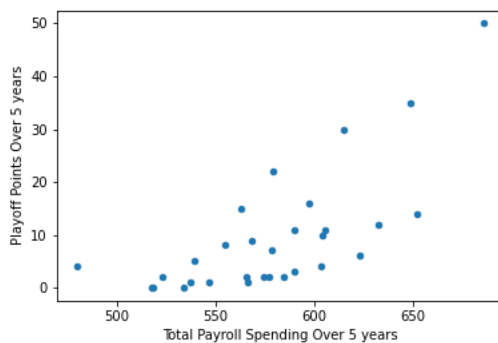
National Basketball Association

Data Table

	Total Payroll Spending Over 5 years	Average WPCT Over 5 years	Playoff Points Over 5 years
Atlanta Hawks	479.89	0.408	4
Boston Celtics	579.14	0.616	22
Brooklyn Nets	584.74	0.45	2
Charlotte Hornets	546.63	0.433	1
Chicago Bulls	537.15	0.373	1
Cleveland Cavaliers	648.82	0.412	35
Dallas Mavericks	523.13	0.451	2
Denver Nuggets	568.15	0.598	9
Detroit Pistons	574.39	0.402	2
Golden State Warriors	685.84	0.598	50
Houston Rockets	597.23	0.591	16
Indiana Pacers	539.38	0.554	5
Los Angeles Clippers	623.11	0.611	6
Los Angeles Lakers	563.15	0.502	15
Memphis Grizzlies	577.41	0.438	2
Miami Heat	652.13	0.534	14
Milwaukee Bucks	605.84	0.637	11
Minnesota Timberwolves	566.1	0.401	1

New Orleans Pelicans	590.25	0.45	3
New York Knicks	533.75	0.365	0
Oklahoma City Thunder	604.1	0.535	10
Orlando Magic	565.36	0.383	2
Philadelphia 76ers	578.45	0.573	7
Phoenix Suns	518.52	0.391	0
Portland Trail Blazers	632.58	0.56	12
Sacramento Kings	517.31	0.411	0
San Antonio Spurs	590.28	0.562	11
Toronto Raptors	614.91	0.632	30
Utah Jazz	554.62	0.63	8
Washington Wizards	603.9	0.466	4

Graphs



Pearson's Correlation

	Playoff Points	WPCT
Payroll Spending	0.708	0.496

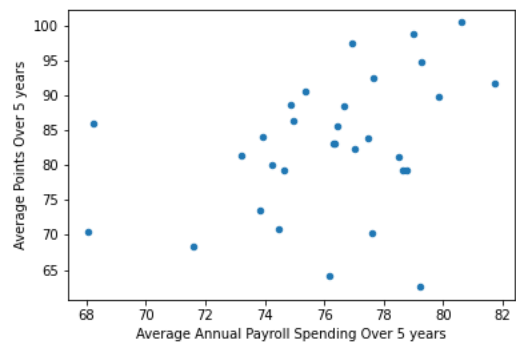
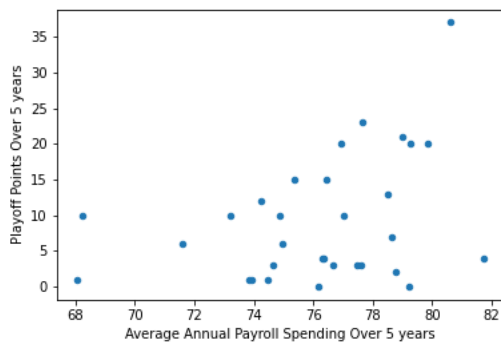
National Hockey League

Data Table

	Average Annual Payroll Spending Over 5 years	Average Points Over 5 years	Playoff Points Over 5 years
Anaheim Ducks	78.64	79.2	7
Arizona Coyotes	74.48	70.8	1
Boston Bruins	76.92	97.4	20
Buffalo Sabres	76.15	64.2	0

Calgary Flames	77.47	83.8	3
Carolina Hurricanes	68.22	86	10
Chicago Blackhawks	78.77	79.2	2
Colorado Avalanche	73.23	81.4	10
Columbus Blue Jackets	74.98	86.4	6
Dallas Stars	78.49	81.2	13
Detroit Red Wings	79.22	62.6	0
Edmonton Oilers	76.35	83	4
Florida Panthers	73.95	84	1
Los Angeles Kings	73.85	73.6	1
Minnesota Wild	76.68	88.4	3
Montreal Canadiens	74.24	80	12
Nashville Predators	75.37	90.6	15
New Jersey Devils	68.07	70.4	1
New York Islanders	76.45	85.6	15
New York Rangers	74.66	79.2	3
Ottawa Senators	71.6	68.4	6
Philadelphia Flyers	76.3	83	4
Pittsburgh Penguins	79.26	94.8	20
San Jose Sharks	77.02	82.4	10
St. Louis Blues	79.86	89.8	20
Tampa Bay Lightning	80.59	100.4	37
Toronto Maple Leafs	81.71	91.6	4
Vancouver Canucks	77.6	70.2	3
Vegas Golden Knights	77.64	92.5	23
Washington Capitals	79	98.8	21
Winnipeg Jets	74.86	88.6	10

Graphs



Pearson's Correlation

	Playoff Points	Season Points
Payroll Spending	0.353	0.369

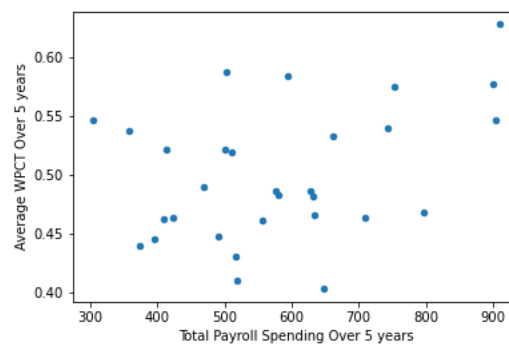
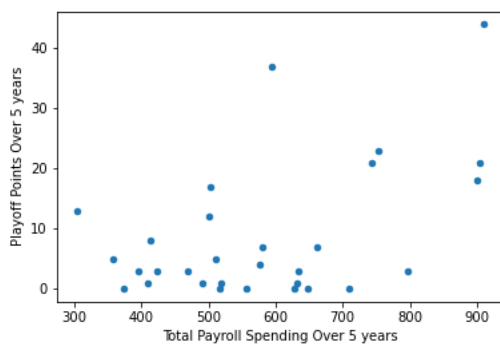
Major League Baseball

Data Table

	Total Payroll Spending Over 5 years	Average WPCT Over 5 years	Playoff Points Over 5 years
Arizona Diamondbacks	468.62	0.49	3
Atlanta Braves	500.25	0.521	12
Baltimore Orioles	518.96	0.41	1
Boston Red Sox	905.02	0.547	21
Chicago Cubs	754.03	0.575	23
Chicago White Sox	409.73	0.462	1
Cincinnati Reds	491.43	0.447	1
Cleveland Indians	502.18	0.587	17
Colorado Rockies	576.83	0.486	4
Detroit Tigers	646.95	0.403	0
Houston Astros	595.02	0.584	37
Kansas City Royals	515.94	0.43	0
Los Angeles Angels	710.09	0.464	0
Los Angeles Dodgers	909.43	0.628	44
Miami Marlins	394.96	0.445	3
Milwaukee Brewers	412.92	0.521	8
Minnesota Twins	510.32	0.519	5
New York Mets	631.87	0.482	1
New York Yankees	900.49	0.577	18

Oakland Athletics	357.39	0.537	5
Philadelphia Phillies	556.59	0.461	0
Pittsburgh Pirates	373.92	0.44	0
San Diego Padres	424.15	0.463	3
San Francisco Giants	796.8	0.468	3
Seattle Mariners	627.27	0.486	0
St. Louis Cardinals	662.59	0.533	7
Tampa Bay Rays	304.52	0.546	13
Texas Rangers	633.92	0.466	3
Toronto Blue Jays	580.64	0.483	7
Washington Nationals	743.69	0.54	21

Graphs



Pearson's Correlation

	Playoff Points	WPCT
Payroll Spending	0.486	0.375

Conclusions

Starting with soccer, it is clear that international football is highly influenced by money. This is perhaps not surprising, as neither Serie A nor EPL have any significant salary caps. The only limitation they face is UEFA's Financial Fair Play Regulation preventing teams from spending more than they earn, which, first, has many loopholes, and second, is ineffective if a team earns a lot in the first place. In all 3 soccer leagues, payroll has a more profound effect on success than transfer spending, with Serie A and EPL having extremely high (>0.8) Pearson's correlation values between payroll and both metrics of success. There are, however, some differences between the European leagues. First, the relationship between playoff points and payroll spending is exponential in the English Premier League, whereas in Serie A, it is linear. The second difference is that Serie A's season points and payroll spending have a somewhat logarithmic relationship, as the positive trend starts leveling off near the right edge of the graph. This suggests that throwing heaps of money on salaries helps only to a certain point, perhaps

because unlike spending, the number of points a team can get is limited. The final difference is that in Serie A, transfer spending is less important for success than in EPL. This is likely because Serie A has more teams exporting players, where many clubs like Atalanta sell their players to stronger clubs and thus have positive transfer balances, while at the same time doing quite well both domestically and in international competitions. In EPL, meanwhile, every single club had a negative transfer balance (i.e. teams spend more than they earn), so the relationship between transfer spending and success is more pronounced. While EPL and Serie A trends have some differences, MLS is entirely unique even though it is the same sport. Payroll spending does have a positive correlation with success, but it is much lower than in either EPL or Serie A. Transfer spending, meanwhile, appears to have virtually no effect on playoff points and, surprisingly, a slight negative correlation with season points. In fact, the scatter plot of transfer spending and season points in MLS appears to have two lines going up and down and forming an X, suggesting that for some teams, generally, the more they spend, the more season points they will get, whereas for others, the opposite may be true. These relationships are very weak, however, so one may assume that transfer spending barely has an effect on regular season performance in MLS, and the variations are either random or caused by other outside factors. The apparent lack of correlation between transfer spending and success may be explained by the implementation of the draft system in MLS, where teams prefer to employ fresh college graduates for free rather than sign expensive players from other teams. Furthermore, the draft system actually favors weaker teams, as brand-new expansion clubs, teams that did not make the playoffs in the previous season, and teams that got knocked out of playoffs earlier will get first picks to make the competition more fair and even. Another factor diminishing the role of transfer spending is the fact that MLS is not a top world league, and truly talented younger players will be reluctant to move to an MLS club regardless of the transfer fee, and will instead prefer to develop in world-class competitions, including Serie A and EPL. The lower influence of payroll spending can also be explained by the above two factors, as well as by the MLS salary cap and salary minimums.

Moving on to other sports, in all of them, the relationship between spending and success is much less pronounced than in European soccer. All of the North American leagues employ certain measures, like salary caps, luxury taxes, and draft systems, that disrupt the natural positive relationship between payroll, regular season performance, and postseason performance. Nevertheless, there is some variation in these relationships across different sports. In the NFL, there is essentially no correlation between any of the metrics. Points on the scatterplots appear to be randomly dispersed, and low Pearson's correlations suggest an extremely weak positive, if not nonexistent, correlation. This can be explained by the fact that out of all North American leagues, the NFL is the strictest when it comes to the salary cap, allowing virtually zero flexibility in exceeding it (save for carryovers from previous seasons). Additionally, football is an extremely coordinated team sport, where individual expensive players may not have as much influence as strategy and teamwork. Other factors unique to football, such as the short NFL season (consisting of only 17 games per team), increased risk of injuries, as well as the playoffs being single elimination, all further increase the role of luck while adding uncertainty to the role of spending on success. Meanwhile, the NBA has the highest correlation between spending and success out of all North American leagues. It has relatively high Pearson's correlation numbers when it comes to both success metrics, but especially playoff

points. Some of the causes for this could include the nature of the sport (individual prowess of highly paid star players like LeBron James and Stephen Curry has a significant impact on success), as well as by NBA's lax salary cap policies (unlike in NFL and NHL, NBA uses a luxury tax and a soft cap with many exceptions, including, among others, the Larry Bird Rule, Rookie Exception, and Mid-Level Exception). Despite implementing a hard cap similar to NFL, NHL has a low but noticeable correlation between spending and success (roughly the same in season points and playoff points). It is still not great, however, and the relationship is certainly not as significant as in NBA or European soccer, likely aided by the aforementioned hard cap. Finally, MLB is the league that has no salary cap at all (though it does use a luxury tax), and its correlation numbers are second only to NBA. However, these numbers were still expected to be higher. In addition to having no cap, MLB is also unique in that each team plays 162 games, by far the highest number out of all the North American leagues. Given the larger sample size, better teams have more opportunities to win and show that they are superior, compared to shorter seasons where a few games have a much larger impact on overall performance. Additionally, baseball is a more individual sport, consisting of a series of one-on-one matchups between a pitcher and a hitter, which would in theory increase the role of highly paid players. Despite these factors, MLB does not have an overly strong relationship between spending and success, but it does have a moderate one.

Overall, in all leagues but NFL, payroll and points/WPCT, as well as payroll and playoff points, have a statistically significant positive correlation. Although a high payroll is not always sufficient to produce a club capable of winning trophies, it has become an increasingly necessary prerequisite for on-field success. As for transfer spending in soccer, its relationship with success is less conspicuous, albeit still relatively significant in Europe while being nonexistent in MLS. Perhaps most surprising, the correlation between spending and playoff points is higher than that with season points or WPCT in all leagues except for NHL, where the two numbers are very close. Theoretically, the element of luck and randomness in the seeding for knockout tournaments would add uncertainty to the relationship between success and spending, but it did the exact opposite.

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